Lesson 05 Demo 05

Creating a React Application Using the Custom Hook

Objective: To develop a React application using Vite that demonstrates the implementation of a custom Hook for managing a counter, enabling increment and decrement functionalities efficiently

Tools required: Visual Studio Code and Node.js

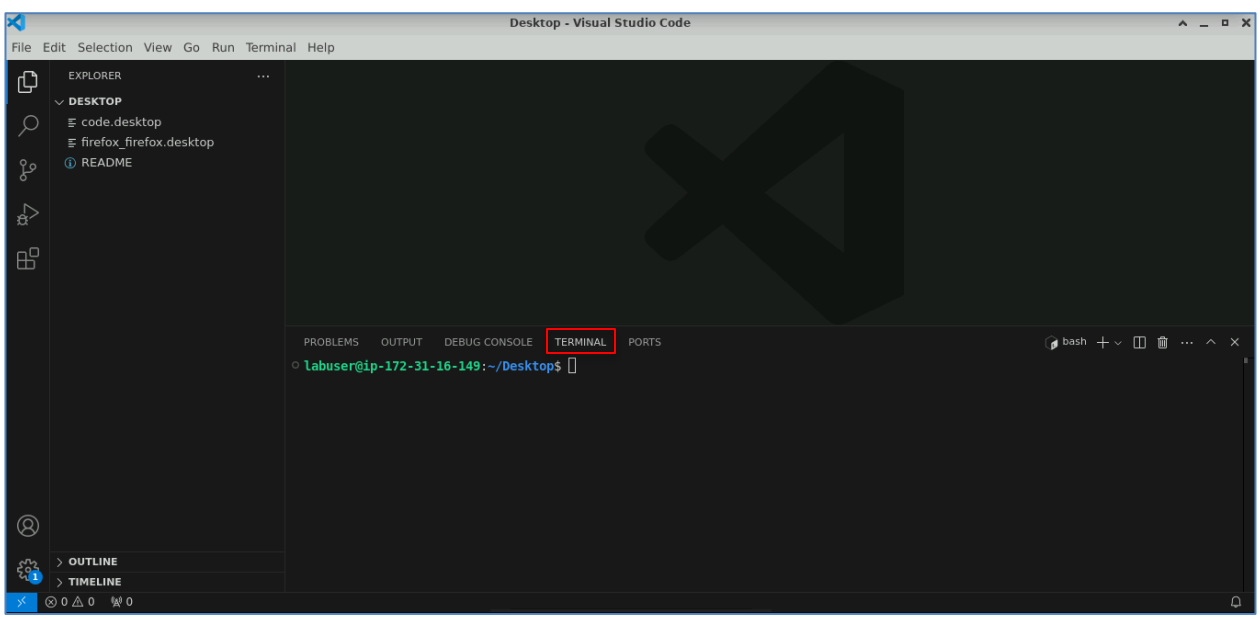
Prerequisites: None

Steps to be followed:

1. Set up a new React project using Vite
2. Create a custom Hook
3. Import and use the custom Hook
4. Run and verify the application

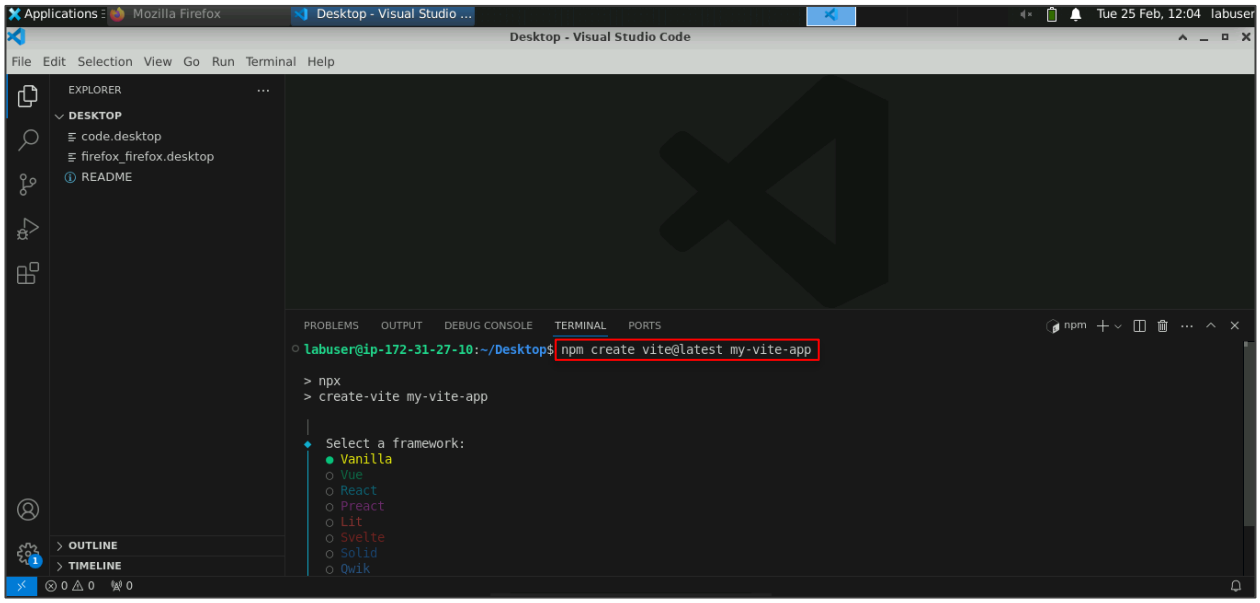
Step 1: Set up a new React project using Vite

1.1 Open Visual Studio Code (VS Code) and launch the terminal

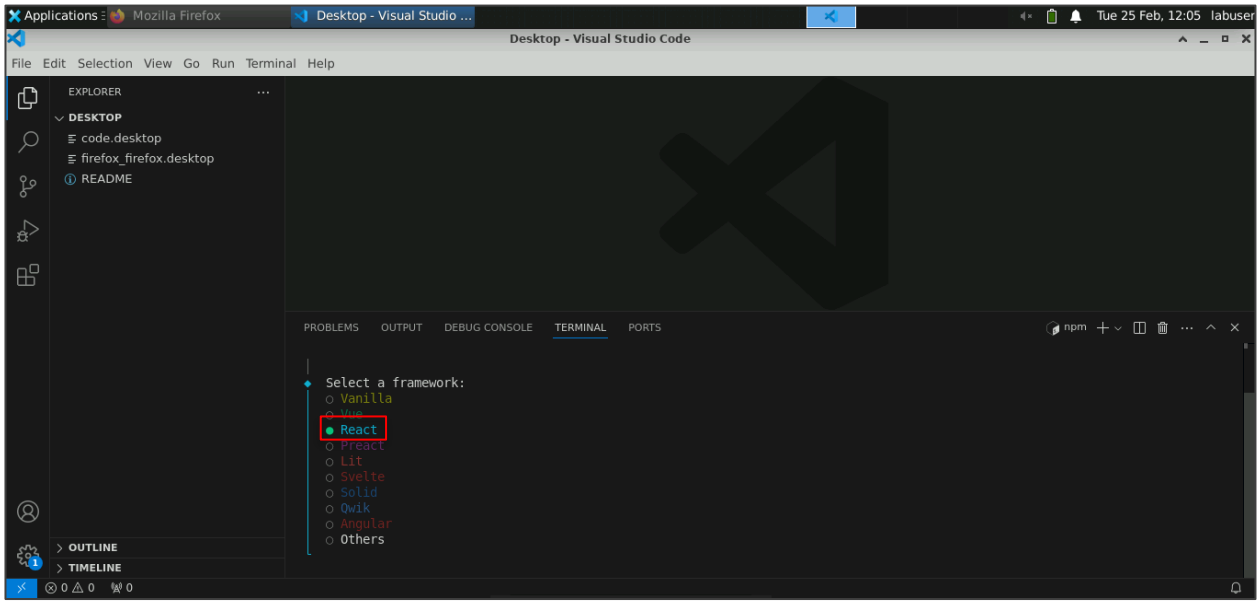


1.2 Run the following command to create a new React project using Vite:

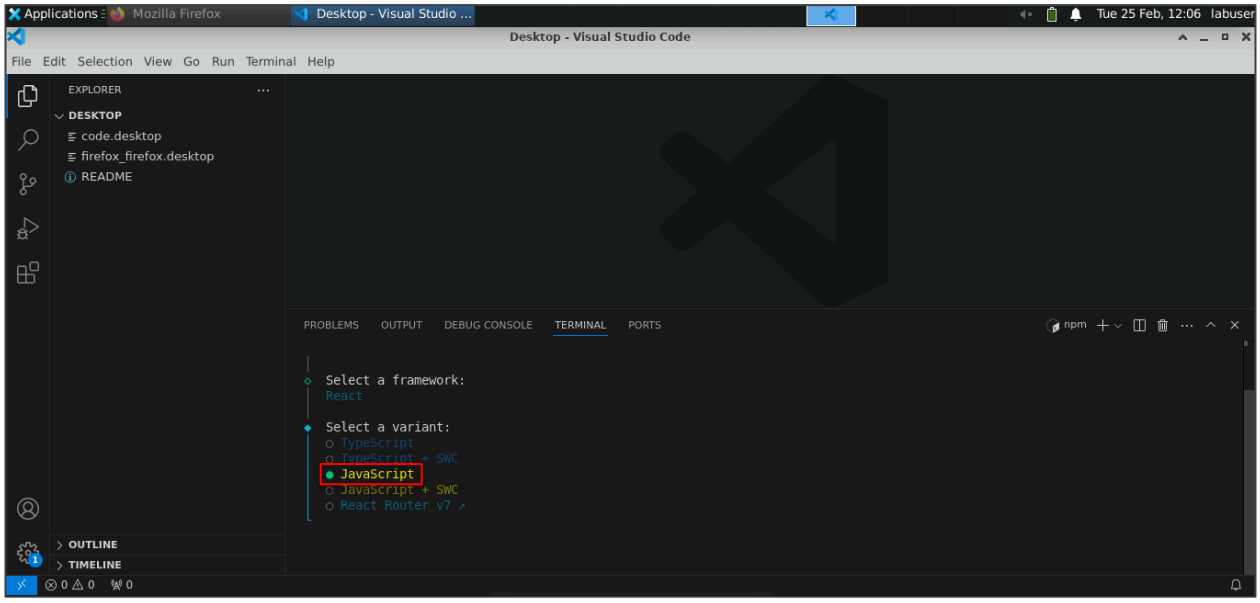
npm create vite@latest my-vite-app



1.3 Select **React** as the project framework and press enter

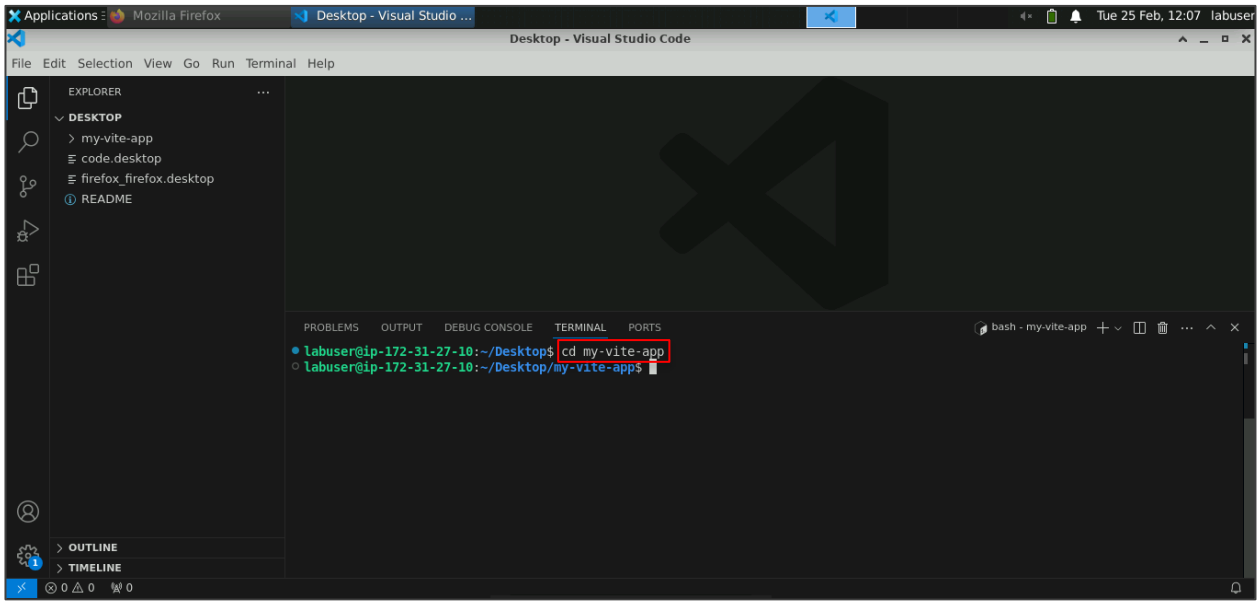


1.4 Select **JavaScript** and press enter



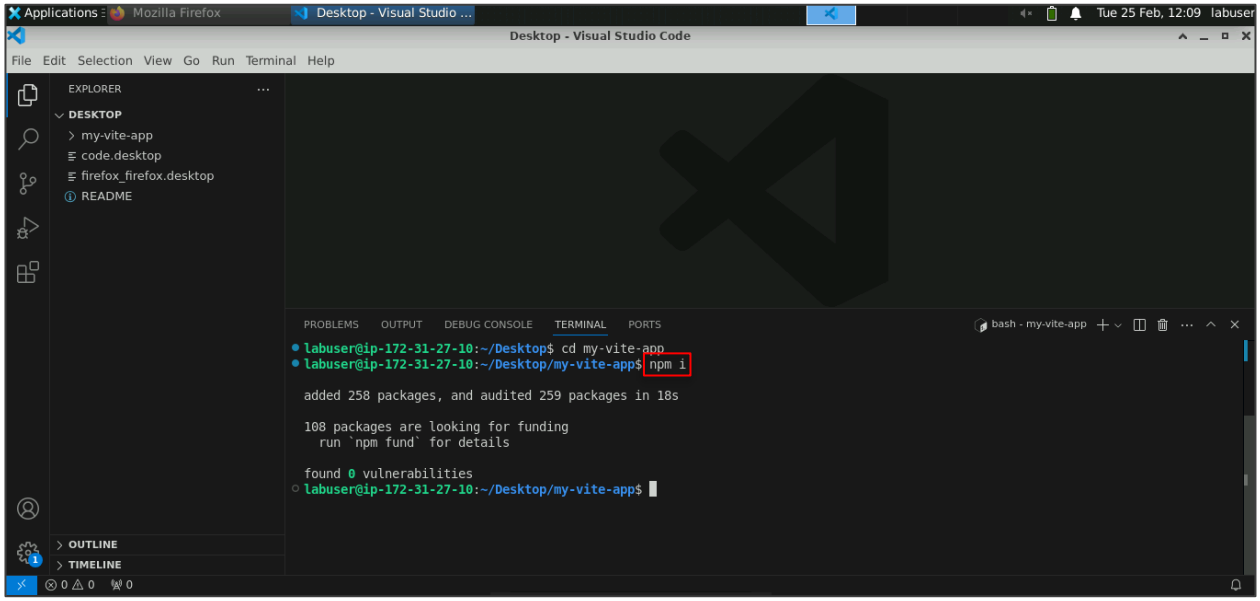
1.5 Run the following command to navigate into the project folder:

cd my-vite-app



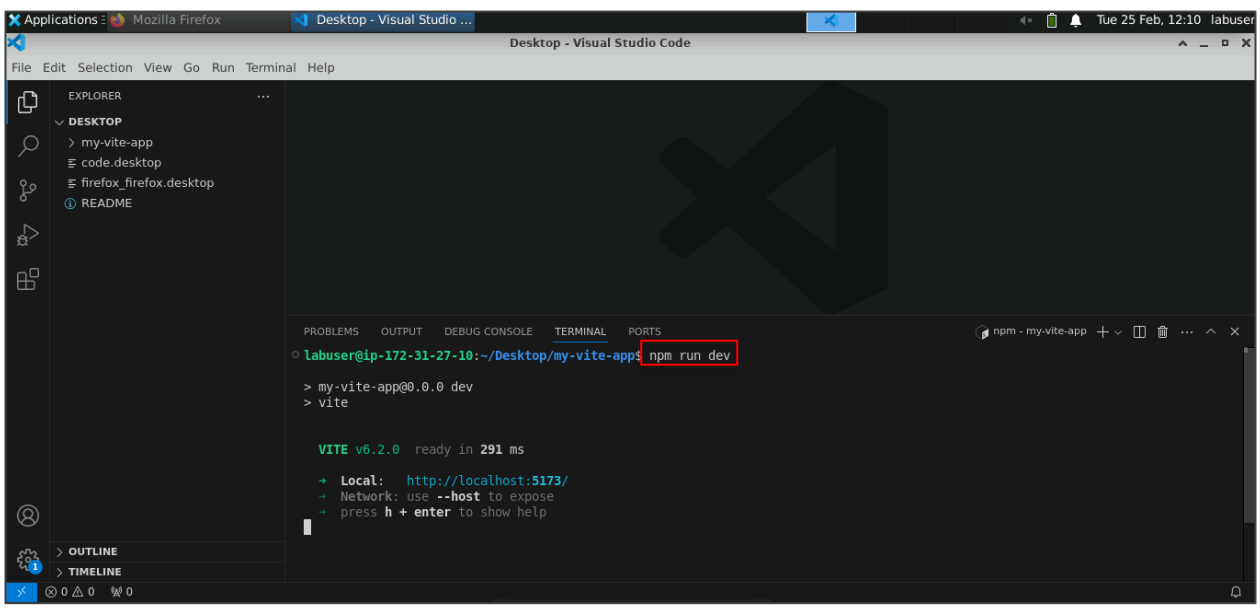
1.6 Run the following command to install dependencies:

npm i

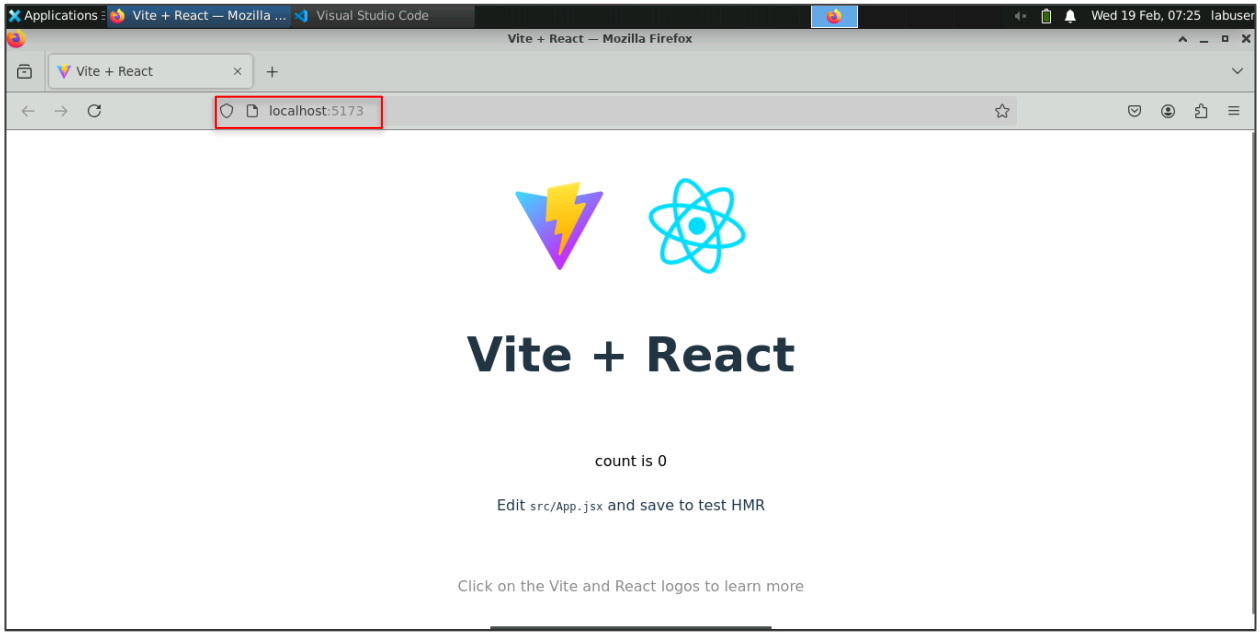


1.7 Run the following command to start the React development server:

npm run dev

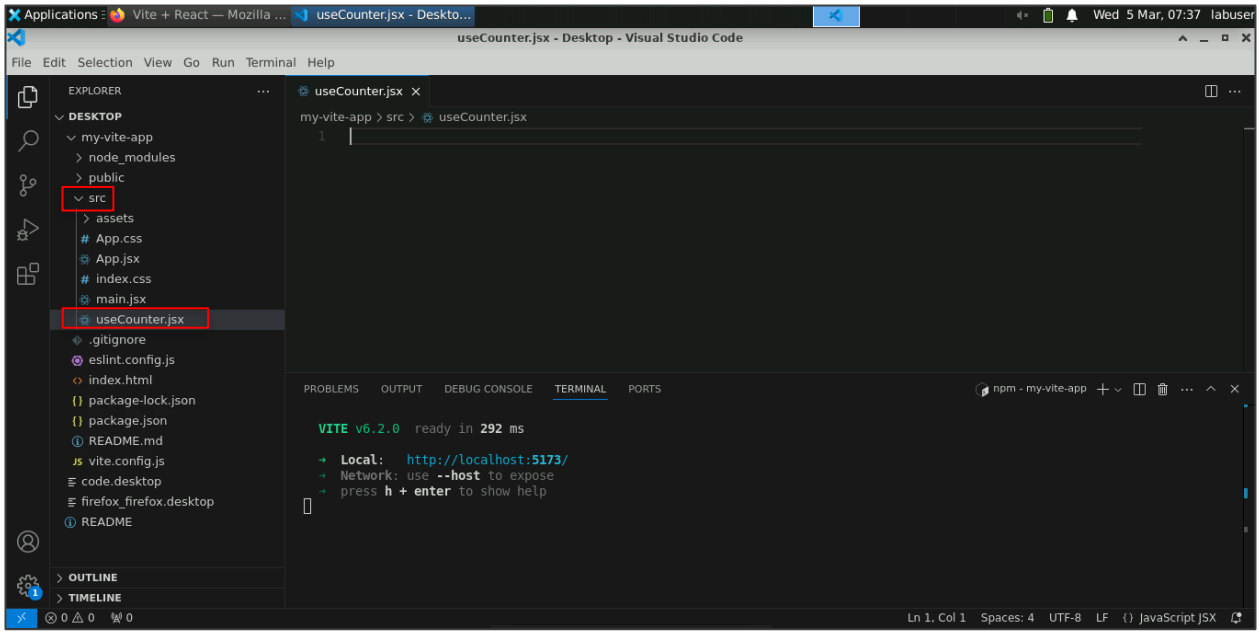


1.8 Navigate to the web browser and paste the following link:
http://localhost:5173



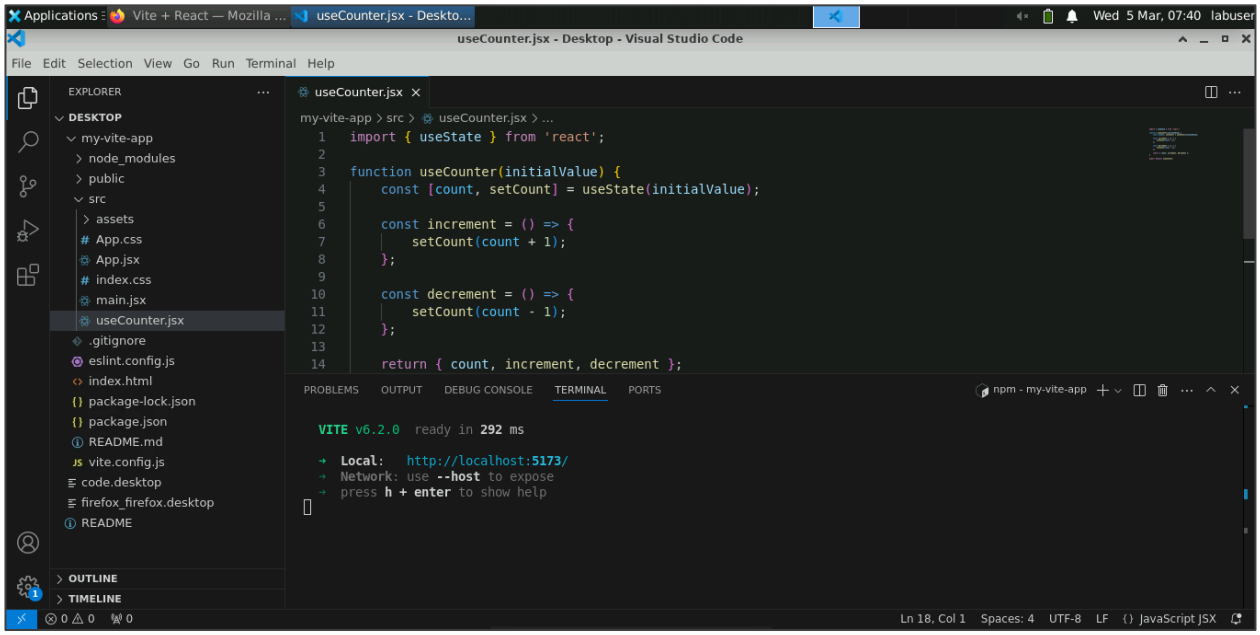
Step 2: Create a custom Hook

2.1 Inside the **src** folder, create a new file named **useCounter.jsx**

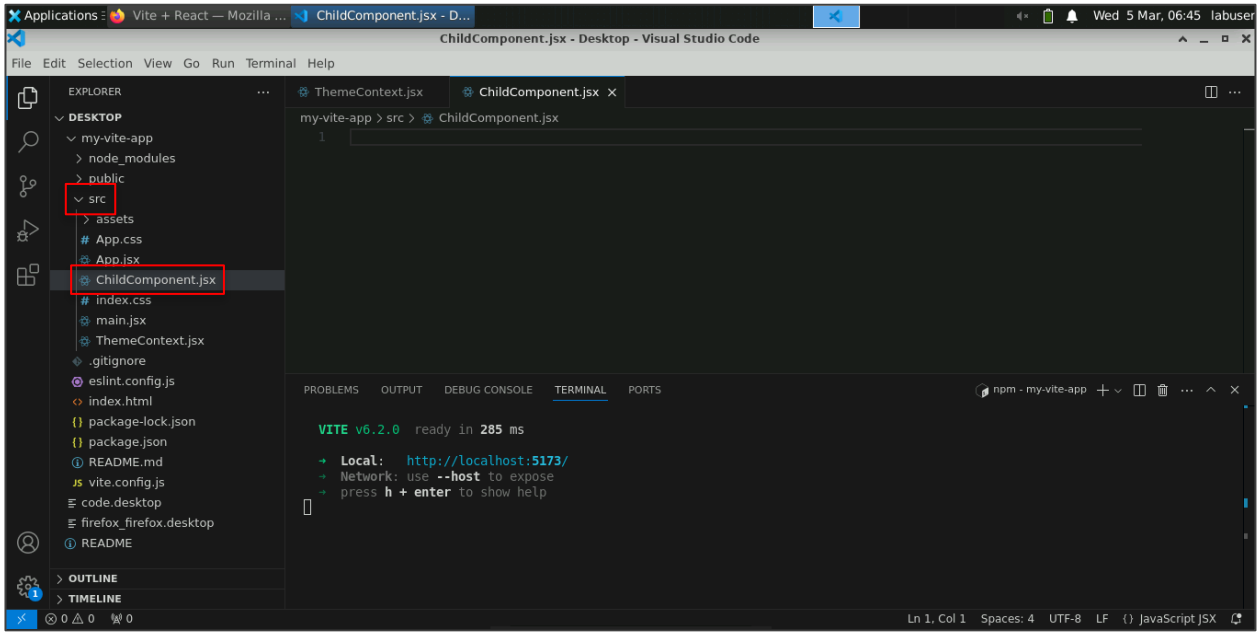


2.2 Add the following code to **useCounter.jsx** to create a custom Hook for managing a counter:

```
import { useState } from 'react';

function useCounter(initialValue) {
  const [count, setCount] = useState(initialValue);

  const increment = () => {
    setCount(count + 1);
  };

  const decrement = () => {
    setCount(count - 1);
  };

  return { count, increment, decrement };
}

export default useCounter;
```

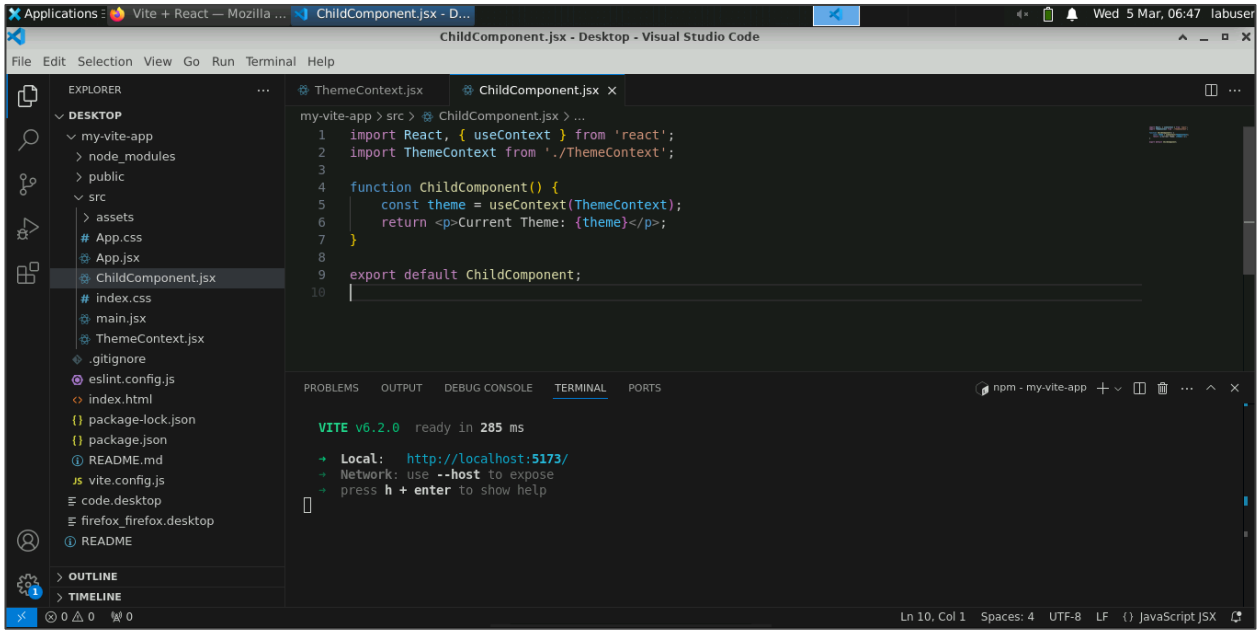


2.3 Inside the **src** folder, create a new file named **ChildComponent.jsx**



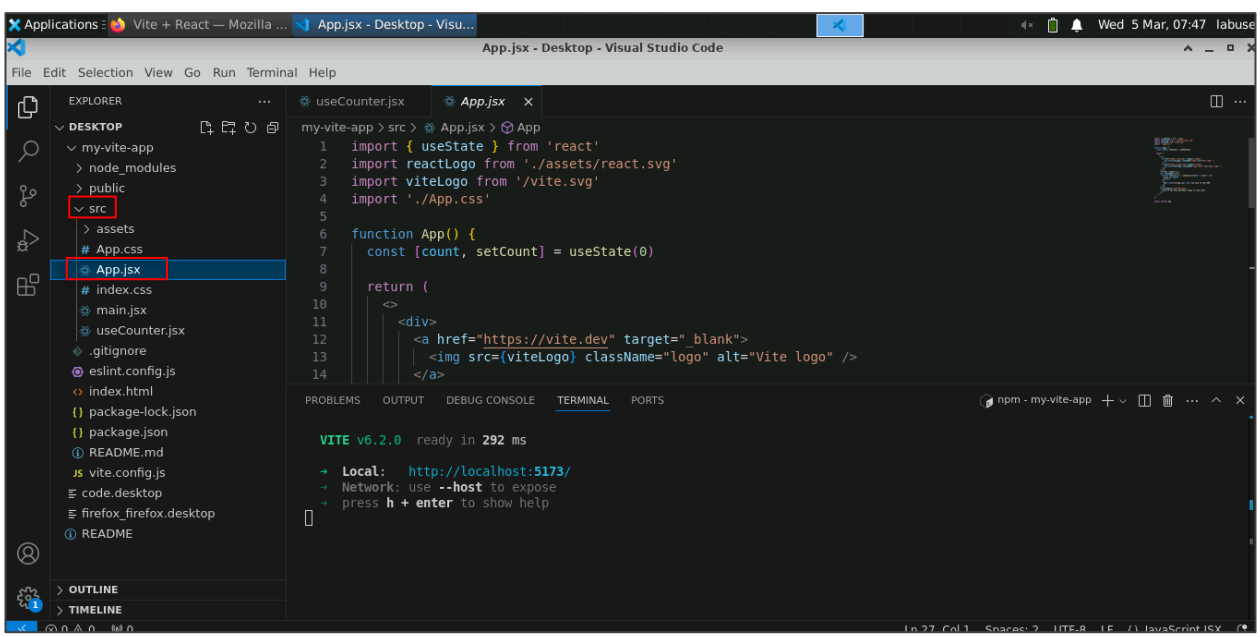
2.4 Add the following code to **ChildComponent.jsx**

```
import React, { useContext } from 'react';
import ThemeContext from './ThemeContext';

function ChildComponent() {
   const theme = useContext(ThemeContext);
   return <p>Current Theme: {theme}</p>;
}

export default ChildComponent;
```



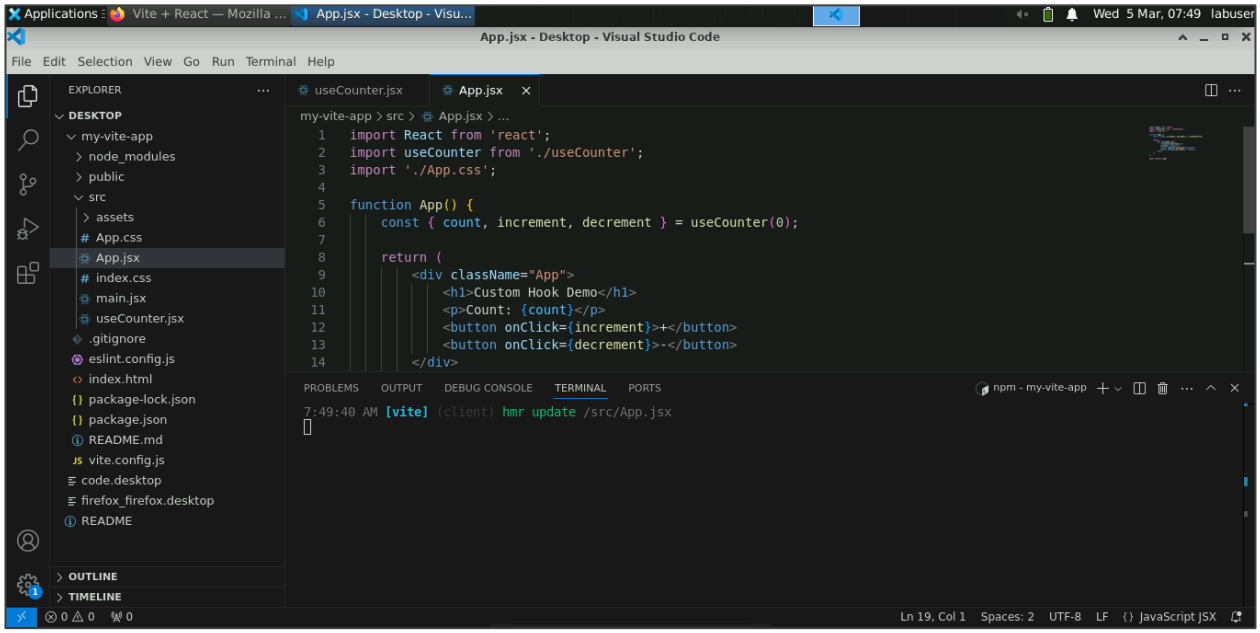
Step 3: Import and use the custom Hook

3.1 Open the **App.jsx** file inside the **src** folder



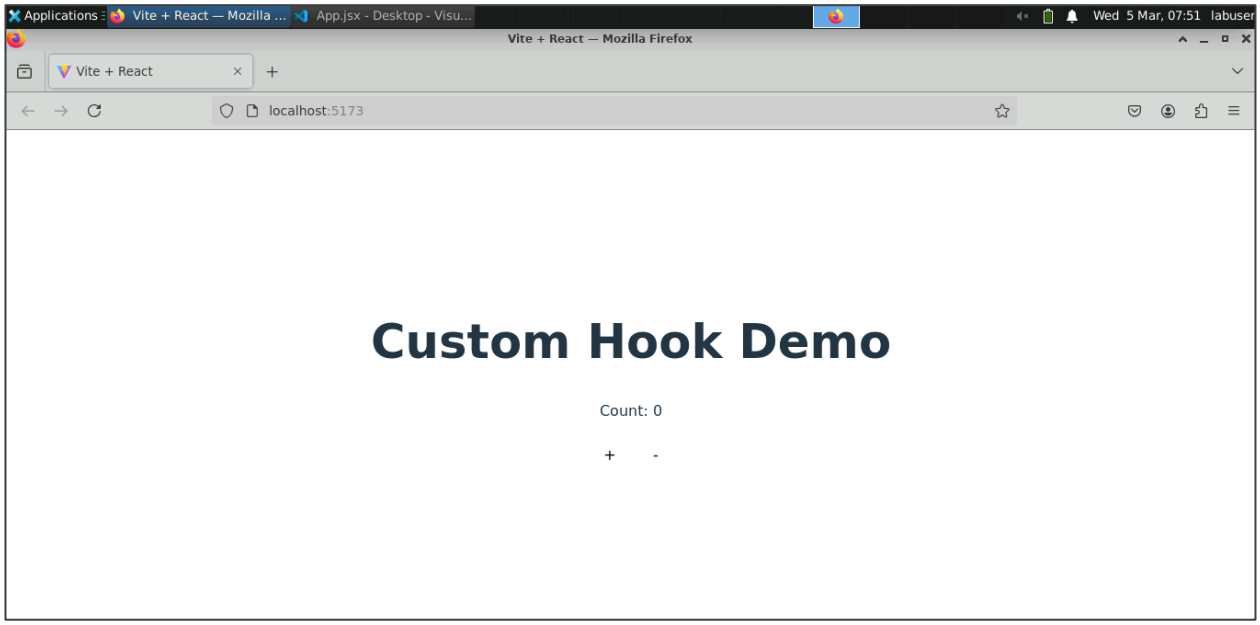
3.2 Paste the following code to include the new components in **App.jsx**:

```
import React from 'react';
import useCounter from './useCounter';
import './App.css';

function App() {
  const { count, increment, decrement } = useCounter(0);

  return (
    <div className="App">
      <h1>Custom Hook Demo</h1>
      <p>Count: {count}</p>
      <button onClick={increment}>+</button>
      <button onClick={decrement}>-</button>
    </div>
  );
}

export default App;
```

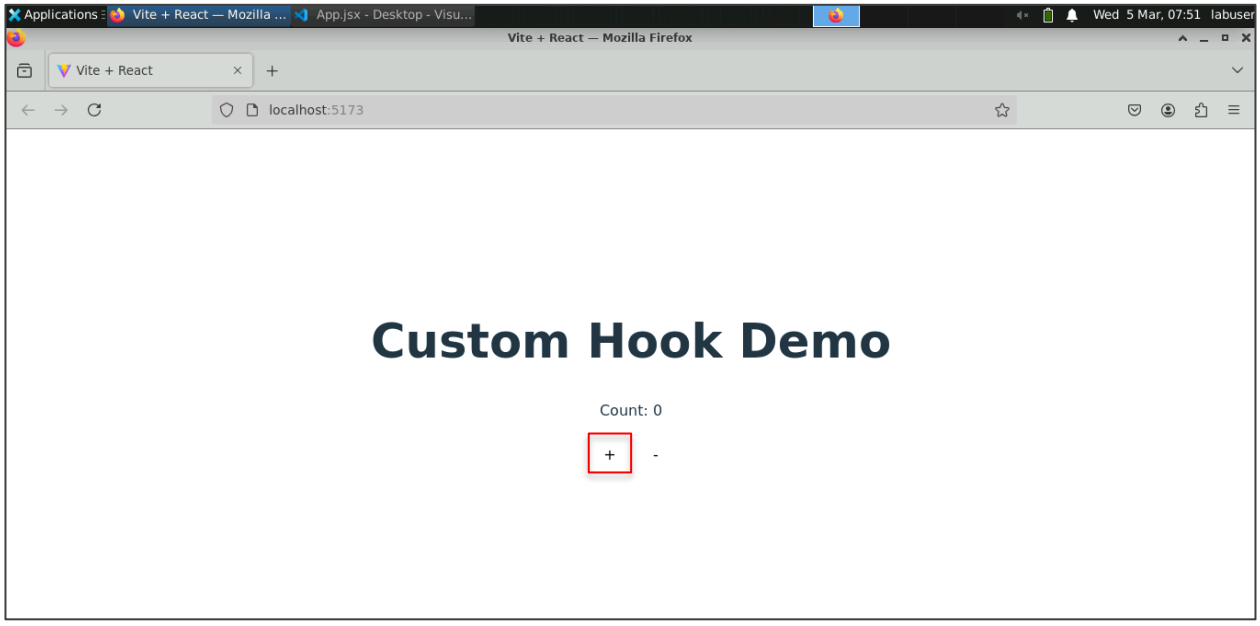


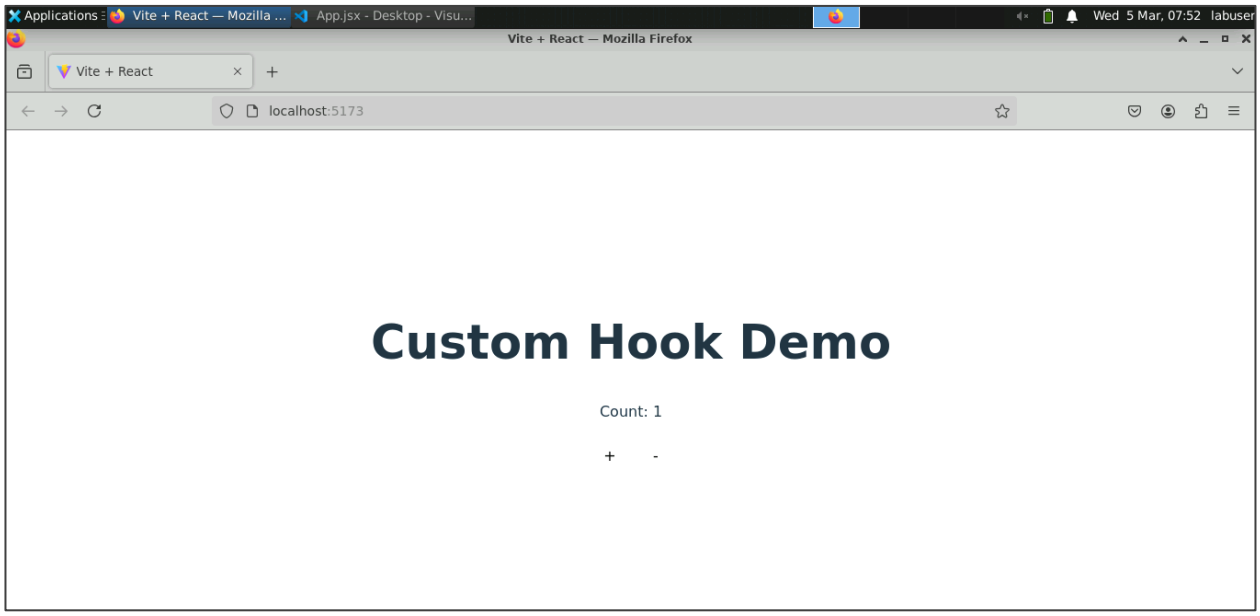
Step 4: Run and verify the application

4.1 Navigate to the browser and refresh the webpage

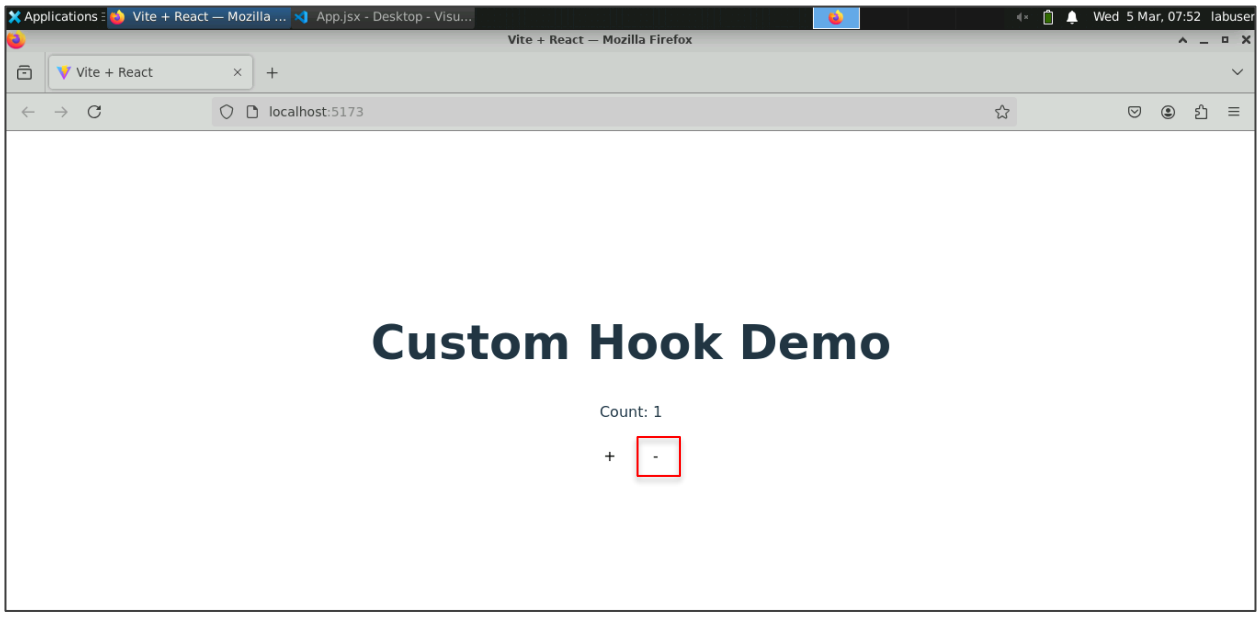


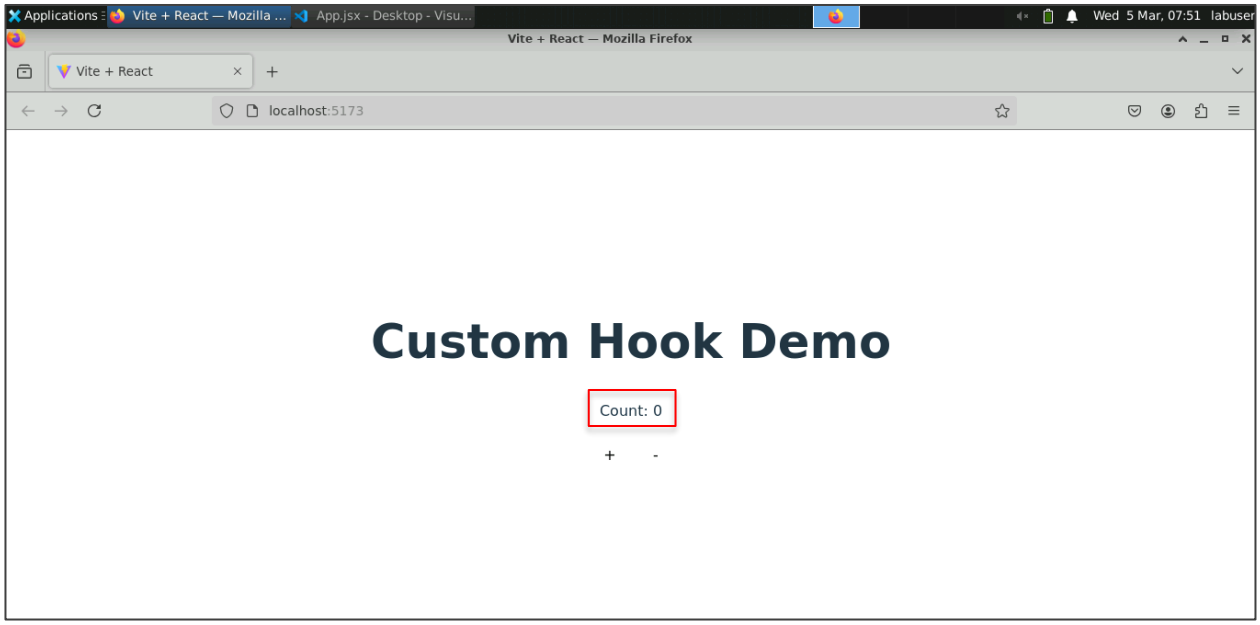
4.2 Click the **+** button to increase the count by 1





4.3 Click the **-** button to decrease the count by 1





By following the above steps, you have successfully created a React application using Vite that demonstrates the implementation of a custom Hook for managing a counter with increment and decrement functionalities, ensuring a faster and more efficient development environment.